

Solution Tutorial VIII – Configurations & Design Process

Which **configurative features** are visible on the three-side view and the illustration of each aircraft?
What are the corresponding **derived properties** for the aircraft? List 10 for each aircraft.

Tasks 1

1) Convair 990: Mach 0.92 long-haul passenger jet of the 1960s

Configurative feature	Derived property
Tube-and-wing configuration	Fuselage cylindrical shape is good for pressurized cabin, well-known configuration (investment cost low)
Four turbojet engines	Redundancy, high total thrust (1960s technology was not as advanced as today's technology)
Anti-shock bodies at wing upper trailing edge	Increase critical Mach number for transonic flight, reduction of wave drag (they could be used as fuel pods as well) → supersonic area rule (<i>rule not part of this lecture</i>)
High wing sweep angle	Lower drag in transonic flight, increase critical Ma number allowing for transonic flight NOT OK: suitable for / enables transonic flight
Low-wing configuration	Easy access for maintenance, uninterrupted payload area
Missing winglets	In the 60s fuel consumption (drag) was not a main design driver. This changed during the oil crisis in the 70s.
Wing-mounted engines	Undisturbed intake flow, reduced root bending moment, access for maintenance
Longitudinal offset of engines	Reduced risk of uncontained engine failure (burst)
Conventional empennage	Less weight, simple actuation system, easy maintenance
Fuselage upsweep	Ground clearance during take-off rotation
Trailing edge device	Low-speed properties

2) *Canadair CL-415: Low speed, amphibious firefighting aircraft*

Configurative feature	Derived property
Large, hydrodynamically shaped hull	Take-off, landing and floating on water
Fuselage step (dt.: Abrisskante)	To allow hydrodynamic lift during take-off = counter the suction force between the water and the hull during take-off
High-wing configuration	Keep water spray away from engine components (corrosion)
Turboprop engines	Suitable for low speeds and low altitude, better SFC than turbofan at low altitude/low speed flight
Wing tip floating devices	Balance during floating on water
Cruciform empennage	Horizontal tail away from engine exhaust, lighter than T-tail, less downwash on horizontal tail than conventional tail
Relatively high aspect ratio of 8	Less induced drag, low-altitude gust sensibility
Semi-retractable landing gear	Main gear can be pulled up on outside of fuselage only → no need for a compartment on inside of fuselage that is waterproof, but still reduced drag and possibility to move on water; also no space in fuselage due to water-payload compartment Nose gear completely retractable → reduced zero-lift drag and possibility to move on water
Wing fences	Diminished spanwise flow
“Finlets” on horizontal tail	Supplement vertical tail and rudder
Winglets	Induced drag reduction

3) Learjet 60: Mach 0.74 medium-haul business jet

Configurative feature	Derived property
Aft-fuselage mounted engines or Engines aft of fuselage (NOT OK: engines at tail or in rear end)	Lower cabin noise, lower positioning of wings possible, less side force asymmetry at OEI, aerodynamically clean wing NOT OK: lower noise
Short landing gear	Autonomous boarding at any airport, lighter than longer landing gear
Retractable landing gear	Reduced zero-lift drag during cruise (compared to non-retractable) NOT OK: reduced drag
Winglets	Induced drag reduction while still fitting in hangar
Wing longitudinal position quite far aft	Due to the engines mounted aft the center of gravity (CG) of the configuration is aft → for longitudinal stability having the CG ahead of the neutral point is required → wing longitudinal position is further aft
Low-wing configuration	Passenger access, payload area uninterrupted
Turbofan engines (NOT OK: turbojet → own mistake if unknown information chosen)	Thrust and SFC suitable for Mach 0.74, reduced time-to-climb
T-Tail configuration	Horizontal tail away from engine exhaust, end-plate effect on vertical tail, less downwash from wings
Ventral fin	Improved stall performance and directional stability at high angles of attack
Trailing edge device	Low-speed properties
Moderate wing sweep angle	Lower drag in lower transsonic flight, increase critical Ma number allowing for transonic flight (lighter than with higher sweep) NOT OK: suitable for / enables transonic flight
Higher sweep in horizontal tail than in main wings	Horizontal tail stalls after main wing; critical Mach number of horizontal tail is higher to avoid loss of elevator effectiveness